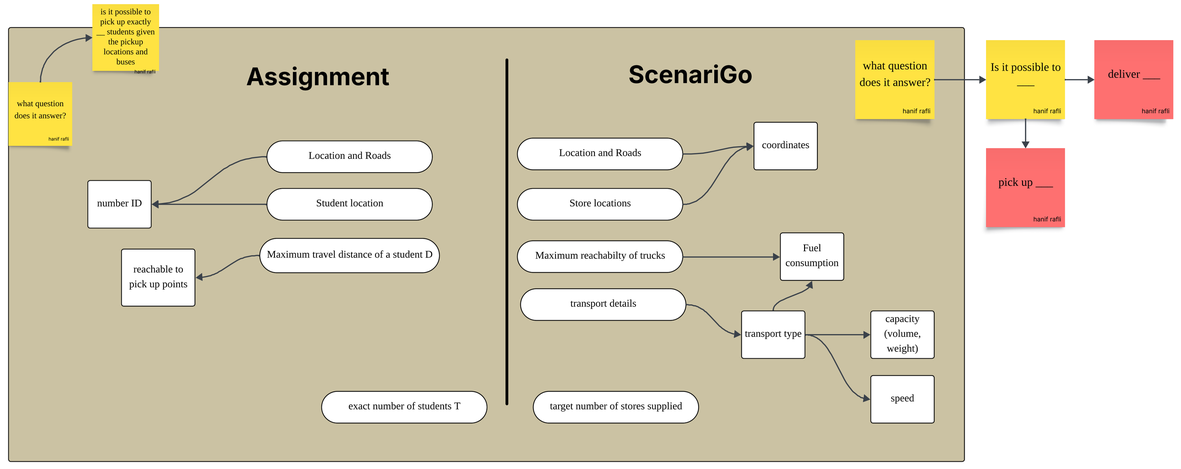
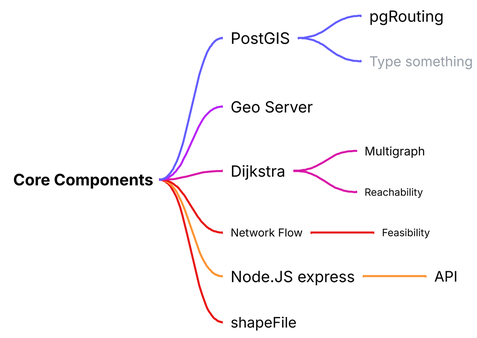
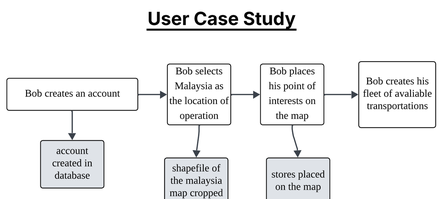




User Case Study



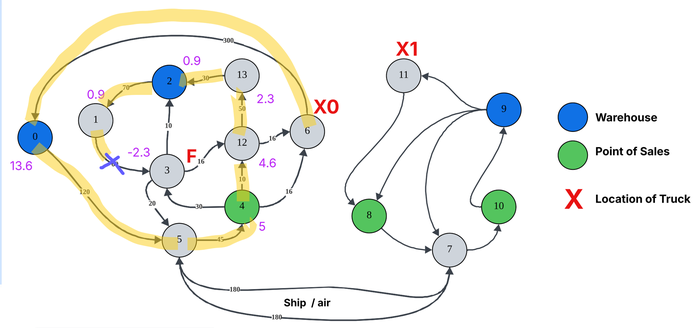
User case study brief:
Bob runs a stationery company in KL, Malaysia. Conveniently, his factories are also warehouses where newly-made pens are stored. They also distribute these stationeries to several stores, supermarkets, etc. across the country.

Additionally, the company owns a number of lorries and each of them is stationed at a certain warehouse. A warehouse may station more than one lorry or none.

Bob also partnered with a carrier company, enabling them to ship/fly their stationery to East Malaysia. For now, Bob wants to keep his shipping domestic (no shipment to other countries are enabled as of now).

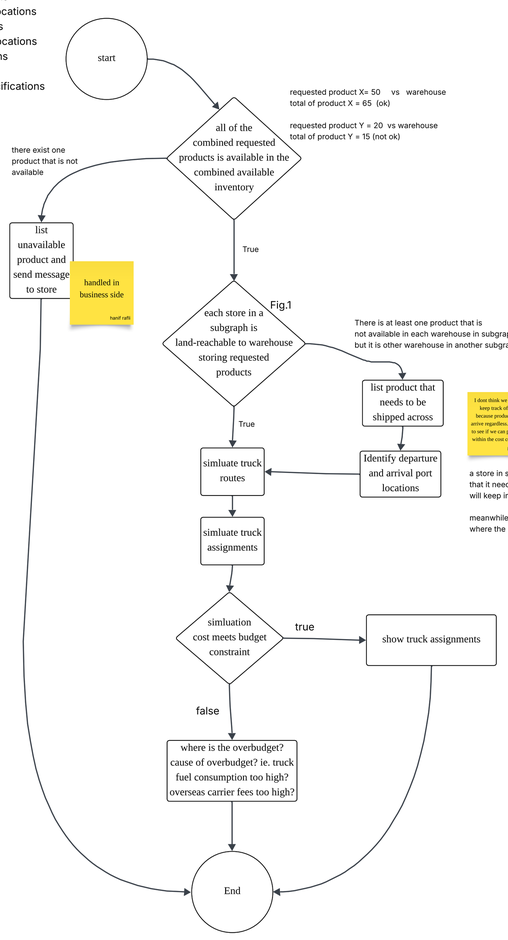
Objective:
Having a number of factories and stores to distribute to, bob would like to know if he could maximize distribution given a configuration .

hanif rafli



Input:

- locations
- truck locations
- gas station locations
- port locations
- warehouse locations
- store locations
- fuel price
- truck(s) specifications



requested product X= 50 vs warehouse total of product X = 65 (ok)

requested product Y = 20 vs warehouse total of product Y = 15 (not ok)

I dont think we need to keep track of time because products will arrive regardless. we want to see if we can pick it up within the cost constraint
hanif rafli

a store in subgraph 1 reports to warehouses in subgraph 2 that it needs product X to be imported. The trucks in subgraph 2 will keep in mind of port destinations to drop off for carrier departure.

meanwhile, trucks in subgraph 1 will keep in mind of which port locations where the products will arrive.

Warehouse locations: [0, 2, 9]

store locations: [4, 8, 10]

truck locations: [6,11]

truck specifications:

[**Tata Ace Gold**
Max weight: 900 kg
Fuel Consumption: 21-22 KMPL
Fuel Tank: 26 L

, **Tata Ace Gold**
Max weight: 900 kg
Fuel Consumption: 21-22 KMPL
Fuel Tank: 26 L]

NAIVE APPROACH
from location 6 to 0: 300 km. I have 26 L of fuel. i will travel 22KM/L. 300 km can be divided into 13.6 sections of 22 KM per Liter. I have used 13.6 Liter. upon arrival at location 0, i have 12.4 L.

From location 0 to 5: 120 KM. I have 12.4 L of fuel. With the same upperbound fuel consumption rate, 120 KM can be divided into 5.4 parts of that consumption. I will have used 5.4 L. upon arrival at location 5, I will have 7 L.

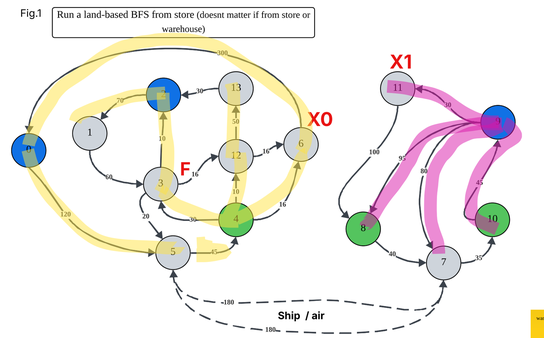
Important: at location 4, I have the choice to go to location 3,12, or 6. The shortest distance here is to 12 (10km). But taking this route will lead me to run out of fuel when i am later enroute to location 3 from 1.

How to decide when to refuel?

simulation will reveal if warehouses/stores are reachable without taking fuel.

if not, a fuel must be taken along the way if discovered.

Fig.1 Run a land-based BFS from store (doesnt matter if from store or warehouse)



observation:
- warehouse location 2, 1 is reachable from store location 4
- warehouse location 9 is reachable from store location 10 and 8

do warehouses in location 2,1 have all products requested by store location 4?

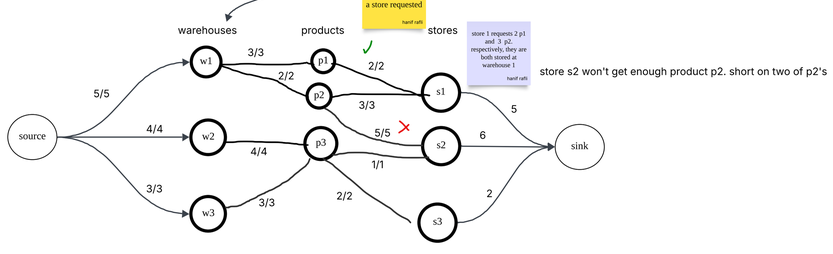
similarly, does warehouse location 9 have all products for stores in location 10 and 8?

warehouse keeps data of which stores are reachable?

warehouse at 0: [4]
warehouse at2: [4]
hanif rafli

[[4,4],[4,4],[8,10]]

determined by which products a store requested
hanif rafli



store s2 won't get enough product p2. short on two of p2's

Products

A4 Paper

Pencil

Warehouses

warehouse 0

warehouse 1

warehouse 2

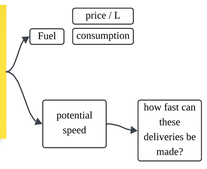
Trucks

Stores

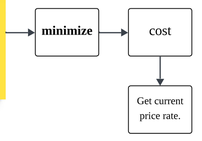
1) BFS determine connection from warehoues to stores

2) connection from products and warehouse determined by which products each warehouse store

Given the MYR 7,000 budget and current locations of my truck, how many stores can i deliver to?
hanif rafli



OPTION: run sim of truck picking up products from warehouses and then distributing to store.
hanif rafli



in island 1, truck 1 has to pick up

Truck 1 has to find the shortest distance to the warehouses that COLLECTS ALL REQUIRED PRODUCTS.

Truck 1 must collect:
- 20 boxes of pens
- 50 boxes of erasers
- 15 boxes of A4 paper

Stores that must receive distribution of the following products:
- **Store 1:** 10 box A4 paper, 20 box of pencils

Consideration:

does a truck have to take the number of total supplies first and then distribute or can it distribute along the way

hanif rafli

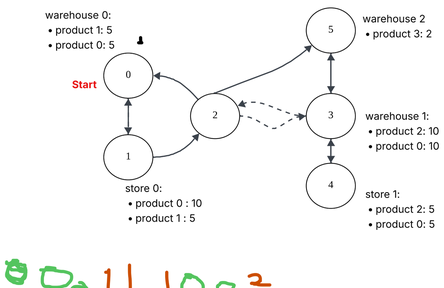
Truck is currently at Sunway

KI warehouse has stock of 5 boxes of A4 paper

Kajang warehouse has stock of 19 boxes of A4 paper

Ipoh warehouse has stock of 5 boxes of A4 paper

- **Greediness solution:**
  - **Take from KL warehouse because it is the nearest one that has stock of A4 paper**
-



[None,None,None,None,None,None,None,None,None,None]

INPUT

FEASIBILITY CHECK 2

This answers which store and which product needs shipping/to be shipped

store 0 need to have product 0 shipped in

warehouse 0 stores 5 of Product-0
hanif rafli

determined by land-reachable vertices That requested the product
hanif rafli

or

distinguish between air and land shipping
hanif rafli

amount of products that the store requested. NOT the same as tallying supply from reachable warehouses.
hanif rafli

edges must be more strict; differentiating between land and via-shipment travels (dotted lines)
hanif rafli

list inludes the warehouses that the shipment comes from

[w0,w1]

HAve to maximize land-routes first. then shipping
hanif rafli

Output:
1. which warehouses are required to fulfill store requests.
2. which warehouses require shipping.
3.

hanif rafli

9/02/2026
The network flow can determine which warehouses the products came from.

To-Do: differentiate land and non-land edges in the network flow

hanif rafli

9/07/2026
Dijkstra instead of BFS. Able to re-use for future reachability questions to other vertices with respect to a particular **warehouse**

hanif rafli

Output:
1. List of locations that must be visited by the trucks
2.

hanif rafli